Pembroke Hill Invitational - Repaired practice edition - Answer key

Division B | Circuit Lab | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

35. (1 points) Authored replacement for source item 35: For a simple solenoid with fixed current and fixed core geometry, does increasing the number of turns per unit length increase or decrease its magnetic field? Explain.

answer: It increases the field; in the ideal long-solenoid model B is proportional to turns per unit length times current. Holding current fixed removes the indirect effect of wire resistance. (1 points).

repair-case. (2 points) Authored written substitute for unavailable source material: Equal LED current guarantees equal brightness?

Unit: LED matching, charge-time estimates, and troubleshooting.
Worked-case data: Estimate ideal time for 1.5 Ah supplied at constant 0.30 A.

A. No

B. Always

C. Only for any red LED

D. Only at 5 V

Correct: A